Hotel Room Reservation System

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MINI LAB PROJECT REPORT

This Report Presented in Partial Fulfillment of the course **CSE114**:

Programming and Problem Solving Lab in the Computer Science and Engineering Department

DAFFODIL INTERNATIONAL UNIVERSITY

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DECLARATION

We hereby declare that this lab project has been done by us under the supervision of Mehadi Hasan, Lecturer, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere as lab projects.

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COURSE & PROGRAM OUTCOME

Course Outcomes (CO):

CO1: Understand and apply structured programming using C.

CO2: Implement file handling techniques for data storage and retrieval.

CO3: Apply modular programming and problem-solving skills in real-world applications.

CO4: Analyze, design, and implement reservation systems ensuring efficiency and accuracy.

Program Outcomes (PO):

PO1: Apply engineering knowledge in designing solutions to practical problems.

PO2: Identify, analyze, and solve computational problems using structured programming.

PO3: Develop effective solutions through modular design and file handling techniques.

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Chapter 1:
Introduction

The first chapter introduces the project, its motivation, and the expected outcomes.

1.1 Introduction

The Hotel Room Reservation System is a C-program-based software solution that automates hotel booking, cancellation, and guest management. It eliminates manual record-keeping and provides a reliable, user-friendly, and efficient way of handling reservations.

1.2 Motivation

Traditional hotel booking involves manual ledgers or spreadsheets, which are prone to errors, overbooking, and delays. Motivated by the need for a digital solution that is accurate, fast, and reliable, this project was developed.

1.3 Objectives

- To implement a user-friendly system for hotel booking and cancellation.
- To provide real-time updates on room availability.
- To store and retrieve guest information using file handling.
- To ensure error handling (date validation, overlapping check-in/out).
- To provide search functionality by guest name or room number.

1.4 Feasibility Study

- **Technical Feasibility:** Implemented in C language with simple file storage; requires minimal resources.
- **Economic Feasibility:** Low-cost solution without database or advanced hardware.
- **Operational Feasibility:** Simple text-based menu-driven interface accessible to all users.

1.5 Gap Analysis

Manual hotel booking is error-prone and existing hotel management software may be too complex or expensive. This project bridges the gap by providing a low-cost, simple, yet efficient reservation system.

1.6 Project Outcome

- Automated reservation and cancellation system.
- Real-time guest information display.
- Data integrity and reduced errors.
- A scalable system that can be expanded to support GUI or online booking.

Chapter 2:
Proposed Methodology/Architecture

2.1 Requirement Analysis & Design Specification

2.1.1 Overview

The Hotel Room Reservation System is a C-based software that manages hotel room booking, cancellation, and guest search. It uses file handling to store and retrieve data, ensuring accuracy and reliability. With a menu-driven console interface, users can easily interact with the system. This project replaces manual booking with a fast, efficient, and user-friendly digital solution.

2.1.2 Proposed Methodology/System Design

The system is modular, consisting of:

1. **Reservation Module** – Books a room with guest details and validates date/time.
2. **Cancellation Module** – Cancels an existing booking and frees the room.
3. **Search Module** – Searches guests by name or room number.
4. **Display Module** – Displays all rooms and booking status.
5. **File Handling Module** – Reads/writes data from rooms.txt.

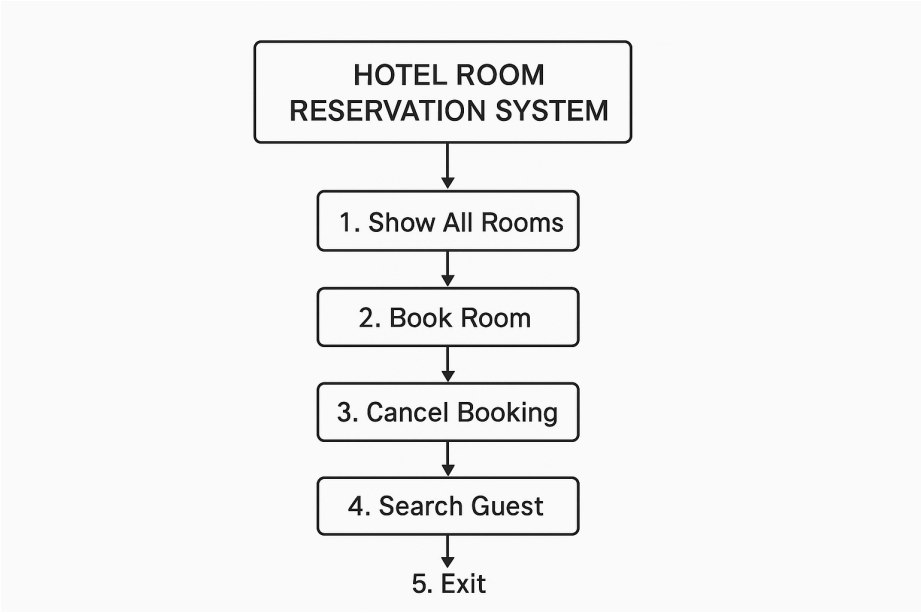

Figure 2.1: **Hotel Room Reservation System** sample diagram**.**

2.1.3 UI Design

```
--- Hotel Reservation Menu ---
1. Show All Rooms
2. Book Room
3. Cancel Booking
4. Search Guest by Name/Room Num
5. Exit
Enter choice:
```

Menu-driven interface.
Options: Show All Rooms, Book Room, Cancel Booking, Search Guest, Exit.
Simple text output with date-time formatting.

2.2 Overall Project Plan

1. **Phase 1:** Requirement Analysis.
2. **Phase 2:** System Design (struct & functions).
3. **Phase 3:** Implementation (modules in C).
4. **Phase 4:** Testing (error handling, date overlap).
5. **Phase 5:** Deployment & Documentation.

Chapter 3:
Implementation and Results

3.1 Implementation

- Implemented in C language using struct Room.
- Date validation (isValidDate), leap year check, and overlapping check.
- File handling (rooms.txt) ensures persistent storage.

Functions:

initializeRooms()
bookRoom()
cancelBooking()
showAllRooms()
searchGuest()

3.2 Performance Analysis

- **Efficiency:** Booking and cancellation execute instantly.
- **Scalability:** Supports 10 rooms but can be extended easily.
- **Accuracy:** Validates dates and prevents double booking.
- **User Experience:** Simple and clear menu interface.

3.3 Results and Discussion

- The system successfully automated manual booking.
- Guest details were stored securely in files.
- Overlapping checks prevented errors.
- The system fulfilled its intended objectives.

Chapter 4:
Engineering Standards and Mapping

4.1 Impact on Society, Environment and Sustainability

- **Impact on Life:** Makes hotel booking faster and more reliable.
- **Impact on Society:** Promotes digital adoption in the hospitality sector.
- **Impact on Environment:** Reduces paper-based records.
- **Ethical Aspects:** Guest data stored securely.
- **Sustainability Plan:** Can be upgraded to an online, database-backed system.

4.2 Project Management and Team Work

- Roles were divided among members (UI, File Handling, Booking Logic).
- Weekly team meetings ensured collaboration.
- Tasks were divided into phases (Design → Code → Test).

4.3 Complex Engineering Problem

4.3.1 Mapping of Program Outcome

- PO1: Applied C programming knowledge to build a real system.
- PO2: Analyzed manual booking problems and provided automated solutions.
- PO3: Developed structured solution with modular design and file handling.

4.3.2 Complex Problem Solving

- Date validation (handling leap years, invalid inputs).
- Overlapping check-in/out detection.
- File synchronization for persistent data.

4.3.3 Engineering Activities

- EA1: Used simple computing resources (text files).
- EA2: User interaction through console menu.
- EA3: Introduced automated hotel management instead of manual ledgers.
- EA4: Improved operational efficiency & reduced waste.
- EA5: Familiarity with C programming, file handling, and modular coding.

Chapter 5:
Conclusion

5.1 Summary

The Hotel Room Reservation System successfully automated booking, cancellation, and guest searching. It reduced errors, saved time, and provided reliable data storage using file handling.

5.2 Limitation

- No graphical interface.
- Limited to 10 rooms.
- File-based storage only (no database).
- No online booking or payment integration.

5.3 Future Work

- Implement a database for scalability.
- Add GUI for better user experience.
- Support online reservations and payments.
- Improve error handling and reporting.

References

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2. E. Balagurusamy – Programming in ANSI C
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